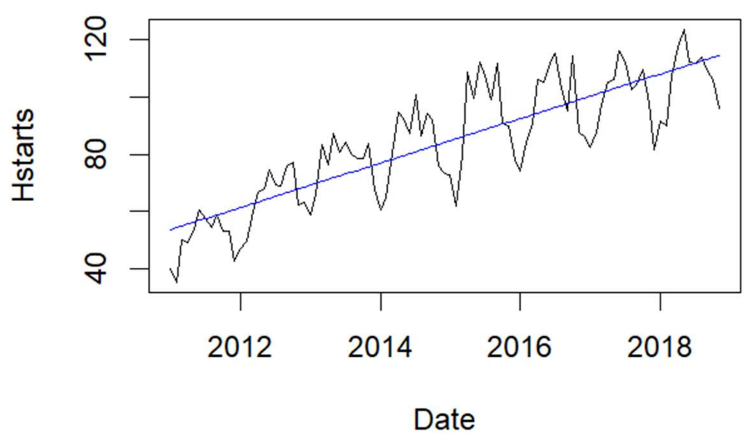
Question 2. Business Scenario

A real estate analytics firm is tracking monthly housing starts to forecast future demand and guide investment decisions.

1. Data Exploration (10 marks)

a. Plot the time series of housing starts.

The graph of monthly housing starts from January 2011 to November 2018 shows clear seasonal patterns. Housing starts are usually higher in the warmer months (spring and summer) and lower in the winter. There is also a clear upward trend, meaning that overall housing demand has been rising over the years.



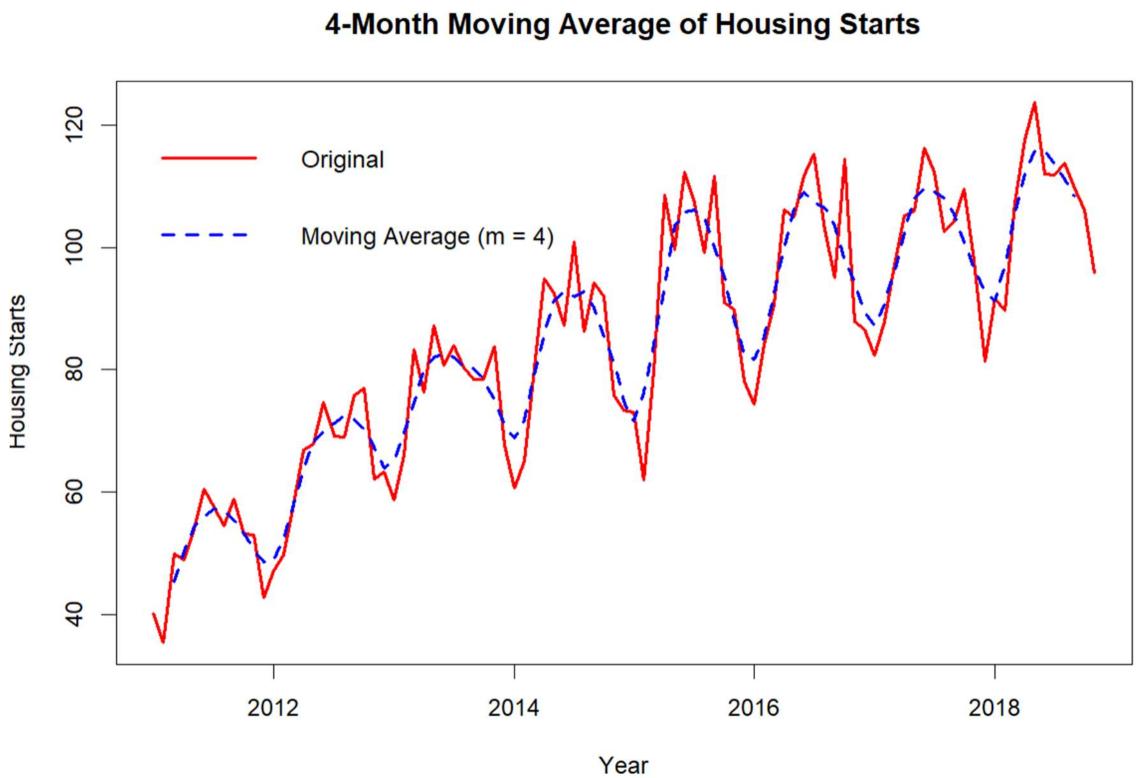
b. What patterns do you observe? Are there any visible trends or seasonal effects?

- **Trend:** Housing starts have steadily increased over the 8 years, showing growing demand for new homes.
- **Seasonality:** The same seasonal pattern happens each year, with the highest activity in the middle of the year and the lowest at the beginning and end.

2. Forecasting Models (60 marks)

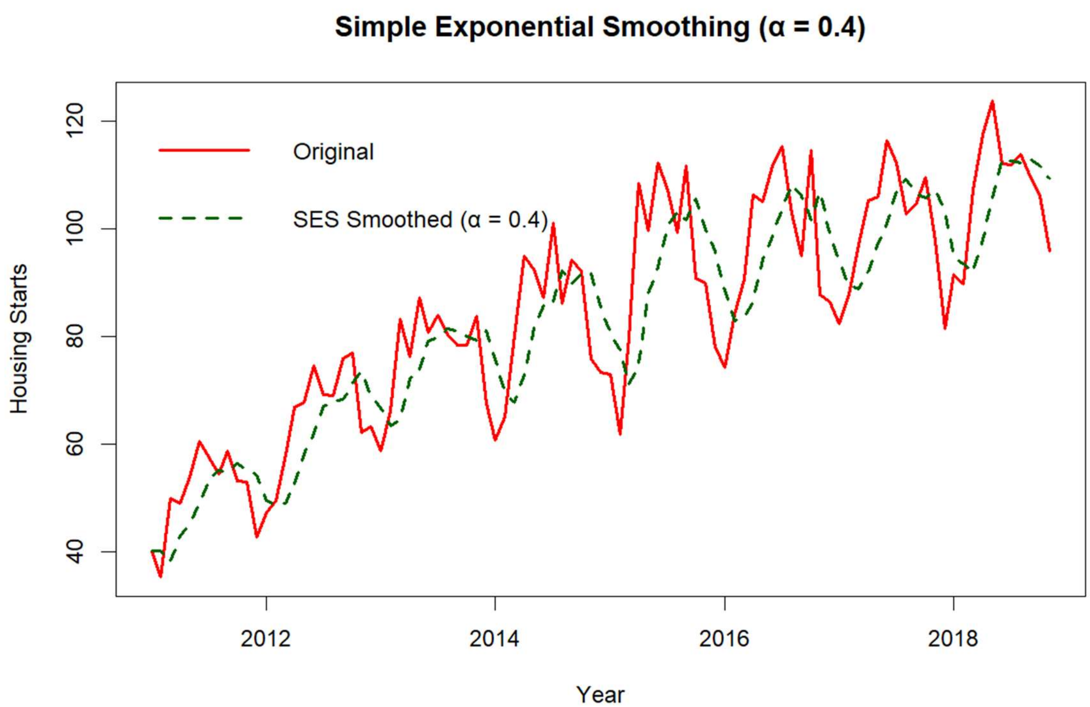
a. Fit a Moving Average model to the data with a sample of m = 4.

A 4-month centered moving average means each value is averaged with nearby months to smooth out short-term ups and downs. This helps reduce noise and makes it easier to see the overall trend in the data.



b. Fit an Exponential Smoothing (ETS) model with an alpha = 0.4.

Simple exponential smoothing with α = 0.4 means recent months have more weight in the forecast, but past data still matters. It smooths out the data to show the overall trend more clearly.



c. Fit a linear trend model.

Captures a straight-line increase or decrease over time. Please refer below image "Time Series Models Fitted to Housing Starts".

d. Fit a seasonal trend model.

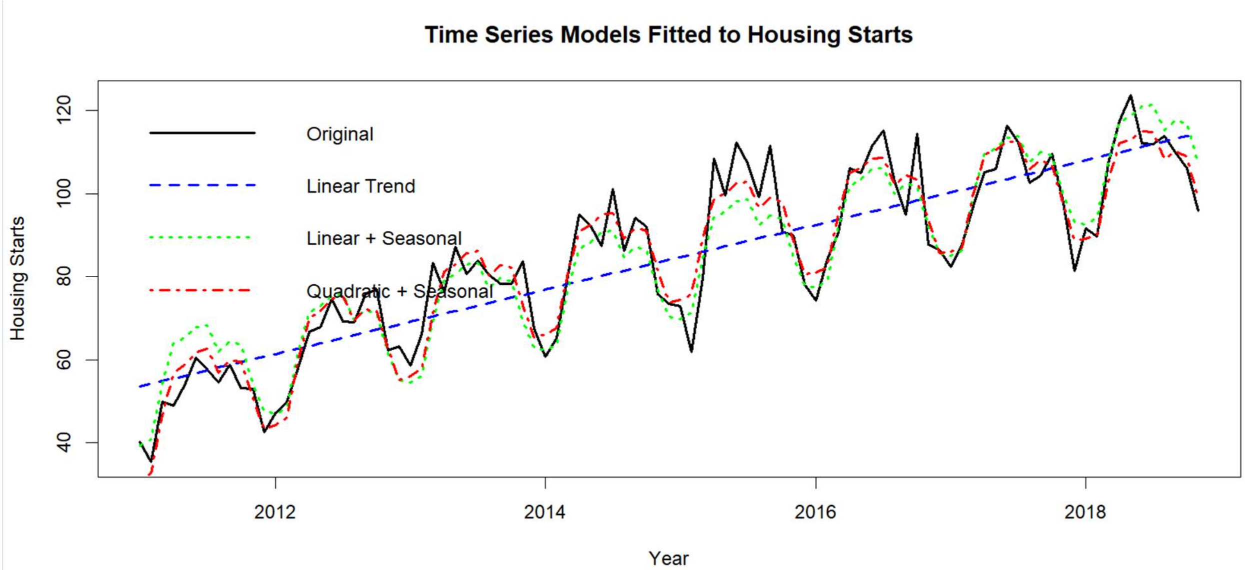
Models predictable repeating changes within each year. Please refer below image "Time Series Models Fitted to Housing Starts".

e. Fit a model with both linear and seasonal trend.

Combines both to better fit real-world data that changes both over time and by season. Please refer below image "Time Series Models Fitted to Housing Starts".

f. Fit a model with both quadratic and seasonal trend

Adds a squared time term to allow for accelerating or decelerating trends (e.g., growth speeding up). Please refer below image



3. Model selection (20 marks)

a. Compare the performance models using RMSE.

We used Root Mean Squared Error (RMSE) to measure how far the predicted values are from the actual data.

The Quadratic + Seasonal model had the lowest RMSE, which means it made the most accurate predictions and best matched the real data.

Exponential smoothing and linear models had higher RMSE, so they weren't as accurate and missed some important patterns in the data.

Output from R Console:

```
> # RMSE for Linear Trend
> rmse_linear <- rmse_base(actual, fitted(linear_model))
>
> # RMSE for Seasonal Model
> rmse_seasonal <- rmse_base(actual, fitted(seasonal_model))
>
> # RMSE for Linear + Seasonal
> rmse_lin_seasonal <- rmse_base(actual, fitted(linear_seasonal_model))
>
> # RMSE for Quadratic + Seasonal
> rmse_quad_seasonal <- rmse_base(actual, fitted(quad_seasonal_model))
>
> # RMSE for Exponential Smoothing
> rmse_ets <- rmse_base(actual, fitted(ets_model))
>
> # Print results
> cat("RMSE for each model:\n")
RMSE for each model:
> cat("1. Moving Average (m=4):      ", round(rmse_ma, 2), "\n")
1. Moving Average (m=4):       5.51
> cat("2. Linear Trend:              ", round(rmse_linear, 2), "\n")
2. Linear Trend:               11.35
> cat("3. Seasonal Only:             ", round(rmse_seasonal, 2), "\n")
3. Seasonal Only:              18.48
> cat("4. Linear + Seasonal:         ", round(rmse_lin_seasonal, 2), "\n")
4. Linear + Seasonal:          6.59
> cat("5. Quadratic + Seasonal:      ", round(rmse_quad_seasonal, 2), "\n")
5. Quadratic + Seasonal:       5.23
> cat("6. Exponential Smoothing:     ", round(rmse_ets, 2), "\n")
6. Exponential Smoothing:      10.85
>
```

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b. Use cross-validation to compare model performance.

Time series cross-validation with rolling forecast origin means we tested the model by making predictions one step at a time, just like in real life when new data keeps coming in.

Consistent with RMSE results means this method also showed that the Quadratic + Seasonal model gave the best predictions.

```
> rmse <- function(errors) sqrt(mean(errors^2, na.rm = TRUE))
>
> # Cross-Validation RMSE for Linear Trend
> cv_linear <- tsCV(housing_ts, function(y, h) forecast(tslm(y ~ trend), h = h), h = 1)
> rmse_linear <- rmse(cv_linear)
>
> # Linear + Seasonal
> cv_lin_seasonal <- tsCV(housing_ts, function(y, h) forecast(tslm(y ~ trend + season), h = h), h = 1)
> rmse_lin_seasonal <- rmse(cv_lin_seasonal)
>
> # Quadratic + Seasonal
> cv_quad_seasonal <- tsCV(housing_ts, function(y, h) forecast(tslm(y ~ trend + I(trend^2) + season), h = h), h =
1)
> rmse_quad_seasonal <- rmse(cv_quad_seasonal)
>
> # Exponential Smoothing (alpha = 0.4)
> cv_ets <- tsCV(housing_ts, function(y, h) forecast(ses(y, alpha = 0.4, initial = "simple"), h = h), h = 1)
> rmse_ets <- rmse(cv_ets)
>
> # Print results
> cat("Cross-Validation RMSEs:\n")
Cross-Validation RMSEs:
> cat("1. Linear Trend:            ", round(rmse_linear, 2), "\n")
1. Linear Trend:             12.13
> cat("2. Linear + Seasonal:       ", round(rmse_lin_seasonal, 2), "\n")
2. Linear + Seasonal:        8.19
> cat("3. Quadratic + Seasonal:    ", round(rmse_quad_seasonal, 2), "\n")
3. Quadratic + Seasonal:     7.37
> cat("4. Exponential Smoothing:   ", round(rmse_ets, 2), "\n")
4. Exponential Smoothing:    10.91
>
```

4. Business Insights (10 marks)

a. What does the forecast suggest about future housing demand?

The forecast shows that housing starts will likely keep increasing, meaning demand for new homes is growing.

Seasonal patterns are expected to continue, with more building activity in spring and summer when construction is usually busiest.

Overall, the housing market is expected to stay strong, which is a good sign for investors and developers planning for future demand.

However, these predictions are based on past trends. Things like changes in the economy, government policies, or supply chain issues could affect what happens.

b. How should the firm adjust its investment strategy based on the forecast?

Based on the forecast, which shows an increase in housing starts and seasonal demand peaks, the firm should consider investing more in areas and periods where demand is expected to grow.

It would be beneficial to increase development projects during the spring and summer months, as these are the busiest times for construction. The firm should also focus on investing in regions that show strong growth in the forecast.

Improving supply chains is another important step to ensure that materials and labor are available when needed, helping to avoid costly delays.

In addition, the firm should remain flexible and closely monitor economic conditions and policy changes, adjusting as needed to reduce potential risks.

Overall, the forecast presents a promising opportunity for growth. By making smart, data-driven decisions about when and where to invest, the firm can position itself for long-term success.

Conclusion

The analysis of housing starts from 2011 to 2018 reveals a clear upward trend and consistent seasonal patterns, with increased activity during spring and summer. Among the forecasting models tested, the Quadratic + Seasonal model provided the most accurate predictions, confirmed by both RMSE and cross-validation results.

The forecasts suggest that housing demand will likely continue to rise, offering a favorable environment for investment. To take advantage of this trend, the firm should increase development efforts during peak seasons, invest in high-growth regions, and strengthen its supply chain.

By staying flexible and monitoring external factors like economic changes or policy shifts, the firm can reduce risks and make informed, data-driven decisions. Overall, the forecast supports a strategic and optimistic outlook for future investments in the housing market.